

A POLYGONAL SPECTRE: STRAIGHT-EDGED STRICTLY CHIRAL APERIODIC MONOTILES

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ABSTRACT. Smith, Myers, Kaplan and Goodman-Strauss showed that the equilateral polygon $\text{Tile}(1, 1)$ is a weakly chiral aperiodic monotile, and that replacing its edges by suitable smooth curves gives the Spectres: shapes that tile the plane, but only non-periodically and only with copies of one handedness, even when reflected copies are allowed. They left open whether piecewise-linear edge replacements can also give Spectres. We show that they can. Replace each edge of $\text{Tile}(1, 1)$ by a triangular bump or dent, alternating around the tile. The result is a polygon with 27 sides, and it is a Spectre for every position of the triangle's peak along the edge, every height below an explicit bound, and all but finitely many such heights at each peak position. So a single polygon can be a strictly chiral aperiodic monotile. The proof is elementary: angle sums and side lengths force every tiling by the polygon to straighten into a tiling by unreflected copies of $\text{Tile}(1, 1)$.

1. INTRODUCTION

An *aperiodic monotile*, or *einstein*, is a shape whose congruent copies tile the plane, but only non-periodically. The question of whether one exists grew out of Wang's work on tilings by coloured squares [Wan61]. Berger [Ber66] found the first aperiodic set of tiles, and Robinson [Rob71] and Penrose [Pen79] reduced the number of shapes to six and then two. Socolar and Taylor [ST11] found a single hexagonal tile with matching rules, which cannot be realised by shape alone as a connected tile. In 2023 Smith, Myers, Kaplan and Goodman-Strauss [SMKG24a] found the *hat*, a polygon that tiles the plane only non-periodically. For a survey of the path to these tiles, see [Kap24].

Every tiling by hats uses reflected copies of the hat as well as unreflected ones [SMKG24a]. This raised the *vampire einstein* question: is there a shape that tiles aperiodically without its reflection? Smith et al. [SMKG24b] distinguish two forms of the question. A tile is *weakly chiral aperiodic* if it tiles the plane with copies of one handedness, and every such tiling is non-periodic, although tilings that mix both hands may be periodic. A tile is *strictly chiral aperiodic* if it tiles the plane, and every tiling by its congruent copies, reflections allowed, uses copies of one handedness only and is non-periodic.

The equilateral polygon $\text{Tile}(1, 1)$, a degenerate member of the hat family, is weakly chiral aperiodic [SMKG24b]. It also tiles periodically when both hands are allowed. Smith et al. then replace the edges of $\text{Tile}(1, 1)$ by copies of a curve, oriented so that a tile and its mirror image can no longer meet. The resulting shapes, the *Spectres*, are strictly chiral aperiodic monotiles [SMKG24b, Theorem 2.2], and their tilings have since been studied in detail; see for example [BGMS26]. Their construction uses a smooth curve, and they note that it extends to

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C^1 curves, which force the vertices of neighbouring tiles to meet. They conclude [SMKG24b, Section 6]:

However, we leave open the question of whether some other paths (e.g., piecewise-linear paths) might permit different tilings, or whether all non-trivial choices of path produce valid Spectres.

A piecewise-linear path adds corners to the tile. A new corner might meet the interior of a side of another tile, and a tiling might then fail to straighten into a tiling by $\text{Tile}(1, 1)$. This paper shows that a triangular path avoids this.

The author’s interactive tiling viewer at <https://drbuild.uk/tiles> shows the geometry. It draws tilings by the hat family, including $\text{Tile}(1, 1)$, and by Spectres with several edge shapes. Its Spectre page, with the *Triangle* edge shape in the *Single* arrangement, draws the tiles $X(p, h)$ studied here, with controls for the peak position p and the height h , and marks the height at which the triangles collide.

Main result. Fix $0 < p < 1$ and $h \neq 0$. Replace each edge of $\text{Tile}(1, 1)$ by the two sides of a triangle whose apex lies at signed height h above the point at fraction p along the edge, and give alternate edges the half-turned triangle. The result is the tile $X(p, h)$ of Section 3. Let φ be the angle of the triangle at its apex, and let

$$\delta = \sqrt{6 - 3\sqrt{3}} \approx 0.8966$$

be the least distance between two edges of $\text{Tile}(1, 1)$ with no common vertex.

Theorem 1.1. *Let $0 < p < 1$ and $0 < |h| < \delta/2$, and suppose that no multiple $j\varphi$ with $1 \leq j \leq 2\pi/\varphi$ is a multiple of $\pi/6$. Then $X(p, h)$ is a polygon with 27 sides and a strictly chiral aperiodic monotile. It is a Spectre in the sense of [SMKG24b, Section 2]. For each p , the condition on φ excludes only finitely many values of $|h|$.*

Theorem 1.1 follows from the more general Theorem 5.1, which replaces the bound on $|h|$ by a condition that bumps do not collide.

Corollary 1.2. *The following polygons are strictly chiral aperiodic monotiles.*

- (1) $X(29/50, 13/50)$, shown in Figures 2 and 4.
- (2) $X(1/2, h)$ for $0 < |h| < \frac{1}{2} \tan \frac{\pi}{12}$ with $|h| \neq \frac{1}{2} \tan \frac{\pi}{24}$.

To our knowledge, this is the first proof that a polygon can be a strictly chiral aperiodic monotile. Several authors have stated that straight-edged Spectres exist, without proof. Tatham [Tat23] suggests a jigsaw-style tab or hole on each edge. Goucher [Gou23] draws an equilateral 28-edge polygonal realisation. Kari’s lecture notes [Kar25, Section 7.4] draw straight-edged bumps and dents, in both the half-turn-symmetric and the alternating forms, and state that the resulting tiles have the Spectre properties. Rust [Rus25] states that any consistent piecewise-linear approximation of a curved Spectre works, for example a small orthogonal shift of the midpoint of each edge. Corollary 1.2(2) proves the last construction for small shifts. Goucher’s polygon is $X(1/2, \sqrt{3}/6)$ up to scale. Its apex angle is $2\pi/3$, so Theorem 1.1 does not apply to it (Remark 7.1).

Idea of the proof. The polygon $X = X(p, h)$ has two kinds of corner: the 13 corners of $\text{Tile}(1, 1)$, whose angles are multiples of $\pi/6$, and 14 apexes with angles φ and $2\pi - \varphi$. The condition on φ keeps the two kinds apart in every tiling, so each bump fits exactly into one dent. The two sides of each triangle have lengths $a = \sqrt{p^2 + h^2}$ and $b = \sqrt{(1-p)^2 + h^2}$, and the alternating arrangement attaches the a -sides to the corners of $\text{Tile}(1, 1)$ whose angles are

multiples of $\pi/2$ and the b -sides to those whose angles are multiples of $2\pi/3$. Two tiles of the same hand then meet along whole edges. Two tiles of opposite hands would lay an a -side along a b -side, and following that side leads to a corner where the angles cannot sum to 2π . So every tiling uses one hand, and it straightens into a tiling by unreflected copies of $\text{Tile}(1,1)$. The results of [SMKG24b] finish the proof.

Organisation. Section 2 fixes terms and collects the results of [SMKG24b] that we use. Section 3 defines $X(p,h)$ and studies its corners. Section 4 proves the two local lemmas, and Section 5 proves the main theorem. Section 6 treats the examples, and Section 7 lists open questions.

2. PRELIMINARIES

2.1. Tilings. A *tiling* by a closed topological disc $X \subset \mathbb{R}^2$ is a set $\mathcal{T}$ of images gX of X under isometries g of the plane, with pairwise disjoint interiors and union $\mathbb{R}^2$. The isometries may reverse orientation. A tile gX is *unreflected* if g preserves orientation and *reflected* otherwise. A tiling is *homochiral* if all its tiles are unreflected, or all are reflected. It is *periodic* if some non-zero translation maps it to itself.

Let X be a polygon. For a tile S of a tiling and a point $x \in S$, the *angle of S at x* is the interior angle of S at x if x is a corner of S , and π if x lies on a side of S but is not a corner.

Lemma 2.1. *Let $\mathcal{T}$ be a tiling by a polygon. Every bounded set meets only finitely many tiles, and every point x has a neighbourhood that meets only the tiles containing x . The angles at x of the tiles containing x sum to 2π .*

Proof. The tiles are congruent, have disjoint interiors and contain discs of a fixed radius, so a bounded set meets finitely many of them. The tiles that do not contain x are then finitely many near x and closed, so they miss a small disc about x . In that disc, each tile containing x covers a sector of angle equal to its angle at x . The sectors have disjoint interiors and cover the disc. See also [GS87, Chapter 3]. $\square$

2.2. The polygon $\text{Tile}(1,1)$. Let $T = \text{Tile}(1,1)$ have vertices $v_0, \dots, v_{13}$ in anticlockwise order, with indices read modulo 14. Table 1 lists the coordinates used in [SMKG24b] and in the reference code for the Spectre, with $s = \sqrt{3}/2$. Edge e_i runs from v_i to v_{i+1} , and every edge has length 1. The edges e_9 and e_{10} are collinear, and v_{10} is a vertex with angle π .

i	v_i	θ_i	i	v_i	θ_i
0	(0, 0)	90°	7	(3, 2)	120°
1	(1, 0)	240°	8	$(3 - s, 3/2)$	270°
2	$(3/2, -s)$	90°	9	$(5/2 - s, 3/2 + s)$	120°
3	$(3/2 + s, 1/2 - s)$	120°	10	$(3/2 - s, 3/2 + s)$	180°
4	$(3/2 + s, 3/2 - s)$	270°	11	$(1/2 - s, 3/2 + s)$	120°
5	$(5/2 + s, 3/2 - s)$	120°	12	$(-s, 3/2)$	90°
6	$(3 + s, 3/2)$	90°	13	(0, 1)	240°

TABLE 1. The vertices v_i of $\text{Tile}(1,1)$ and its interior angles θ_i , with $s = \sqrt{3}/2$.

The angle θ_i is a multiple of $\pi/2$ when i is even and a multiple of $2\pi/3$ when i is odd [SMKG24b, proof of Lemma 2.1]. We call v_i *blue* when i is even and *red* when i is odd. Thus, in anticlockwise order, e_i runs from blue to red when i is even and from red to blue when i is odd. (This

parity of vertices differs from the even and odd tiles and edges of [SMKG24b, Section 3], which refer to directions.)

2.3. Results from [SMKG24b]. We use three facts about $\text{Tile}(1, 1)$.

Theorem 2.2 (Smith, Myers, Kaplan and Goodman-Strauss).

- (1) *Every tiling by $\text{Tile}(1, 1)$ is vertex-to-vertex, with v_{10} counted as a vertex: no vertex of a tile lies in the interior of an edge of another* [SMKG24b, proof of Lemma 2.1].
- (2) *There is a homochiral tiling by $\text{Tile}(1, 1)$. Indeed, a Spectre tiles the plane* [SMKG24b, Theorem 2.2], *and every tiling by a Spectre straightens to a homochiral tiling by $\text{Tile}(1, 1)$* [SMKG24b, Section 2].
- (3) *Every homochiral tiling by $\text{Tile}(1, 1)$ is non-periodic* [SMKG24b, Section 2.1, with the proofs of Sections 3–5].

Lemma 2.3. *In a tiling by $\text{Tile}(1, 1)$, a blue vertex of one tile never coincides with a red vertex of another. So two tiles that share an edge share its blue end and its red end.*

Proof. By Theorem 2.2(1), the tiles at a vertex x of the tiling all have a vertex at x . If the blue angles there sum to $m\pi/2$ and the red ones to $2n\pi/3$, then $3m + 4n = 12$ with $m, n \geq 0$, so $m = 0$ or $n = 0$. The second claim follows, since shared edges have the same end points in both tiles. $\square$

3. THE TILES $X(p, h)$

3.1. Construction. Let J be the quarter turn anticlockwise. For each edge e_i define *edge coordinates* by

$$E_i(u, w) = v_i + u(v_{i+1} - v_i) + wJ(v_{i+1} - v_i).$$

Since T is anticlockwise, $w > 0$ points into T . The half turn about the midpoint of e_i acts by $(u, w) \mapsto (1 - u, -w)$.

Fix $0 < p < 1$ and $h \neq 0$. Let γ be the path $(0, 0), (p, h), (1, 0)$ in the (u, w) -plane, and $\gamma^* = \{(1 - u, -w) : (u, w) \in \gamma\}$ its half turn, the path $(0, 0), (1 - p, -h), (1, 0)$. Put

$$\gamma_i = E_i(\gamma) \quad (i \text{ even}), \quad \gamma_i = E_i(\gamma^*) \quad (i \text{ odd}),$$

let c_i be the apex of γ_i , and let Δ_i be the closed triangle $v_i c_i v_{i+1}$. The apex c_i lies at distance $|h|$ from the line through e_i , and its foot lies inside e_i . We call Δ_i a *dent* if c_i lies on the inner side of e_i , and a *bump* otherwise. For $h > 0$ the dents are the Δ_i with i even; for $h < 0$, those with i odd. Write

$$a = \sqrt{p^2 + h^2}, \quad b = \sqrt{(1-p)^2 + h^2}, \quad \alpha = \arctan \frac{|h|}{p}, \quad \beta = \arctan \frac{|h|}{1-p}, \quad \varphi = \pi - \alpha - \beta.$$

So φ is the angle of each Δ_i at its apex. Figure 1 shows the two paths.

We use two conditions on (p, h) .

Definition 3.1.

- (G) Every dent lies in T and meets ∂T only in its base. Every bump meets T only in its base. The dents are pairwise disjoint, and so are the bumps.
- (N) For every integer j with $1 \leq j \leq 2\pi/\varphi$, the angle $j\varphi$ is not an integer multiple of $\pi/6$.

Assume (G). The sides of the triangles Δ_i , other than their bases, form a closed polygonal curve $\Gamma = \gamma_0 \gamma_1 \cdots \gamma_{13}$. Two triangles of the same kind lie on edges with no common vertex

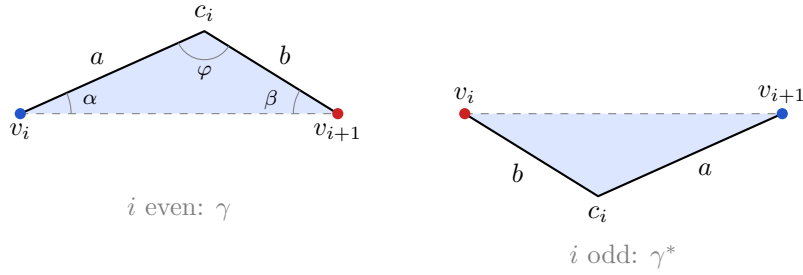


FIGURE 1. The two paths, drawn in edge coordinates with $p = 29/50$ and $h = 13/50$. The inside of the tile is above each edge. Blue marks the vertices with angles in $\{\pi/2, \pi, 3\pi/2\}$, red those with angles in $\{2\pi/3, 4\pi/3\}$. The side of length a always meets a blue vertex.

and are disjoint. A dent Δ_i and a bump Δ_j meet at most in $\Delta_i \cap e_j \subseteq \Delta_i \cap \partial T = e_i$, so in $e_i \cap e_j$. Hence Γ is simple. Let $X = X(p, h)$ be the closed region bounded by Γ . Then

$$X = \left(T \setminus \bigcup_{\text{dents}} (\Delta_i \setminus e_i) \right) \cup \bigcup_{\text{bumps}} \Delta_i.$$

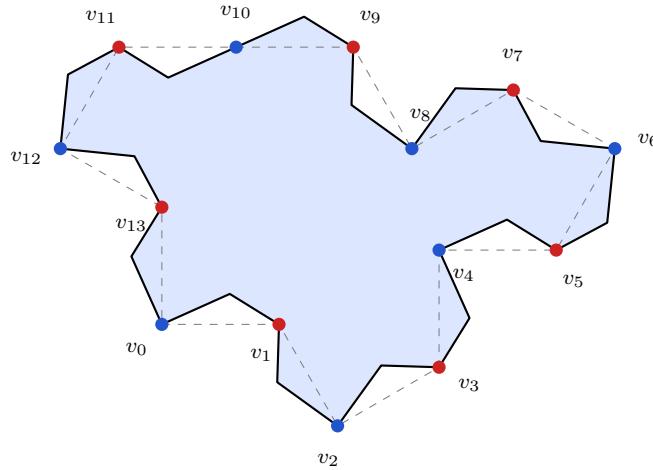


FIGURE 2. The tile $X(29/50, 13/50)$, drawn over $\text{Tile}(1, 1)$ (dashed). Blue dots mark the vertices v_i with i even, red dots those with i odd. The two sides through v_{10} are collinear, so v_{10} is not a corner.

3.2. The two conditions.

Proposition 3.2. *Let δ be the least distance between two edges of T that have no common vertex. Then $\delta = \sqrt{6 - 3\sqrt{3}}$, the distance between v_4 and v_8 , and condition (G) holds whenever $0 < |h| < \delta/2$.*

Proof. The edges of a simple polygon do not cross, so the distance between two disjoint edges is attained at an end point of one of them. A direct computation over the 77 pairs of edges with no common vertex gives $\delta^2 = |v_8 - v_4|^2 = (3/2 - \sqrt{3})^2 + 3/4 = 6 - 3\sqrt{3}$; the ancillary file checks this in exact arithmetic.

Each Δ_i lies within distance $|h|$ of e_i , since its apex is at distance $|h|$ from the line through e_i with foot inside e_i . Since $|h| < \delta$, Δ_i misses every edge that has no common vertex with e_i , and since $2|h| < \delta$, it misses every triangle on such an edge.

Now let e_j share the vertex v with e_i . The triangle Δ_i lies in the sector at v bounded by e_i and the ray from v through c_i . This sector has angle α or β , both less than $\pi/2$. It lies on the inner side of e_i for a dent and the outer side for a bump. The edge e_j makes the angle $\theta_v \geq \pi/2$ with e_i on the inner side, and $2\pi - \theta_v \geq \pi/2$ on the outer side, since $\pi/2 \leq \theta_v \leq 3\pi/2$. So $\Delta_i \cap e_j = \{v\}$.

Hence $\Delta_i \cap \partial T = e_i$. The set $\Delta_i \setminus e_i$ is connected, misses ∂T , and contains points near the midpoint of e_i on the side of c_i . So it lies in the interior of T for a dent and in the exterior for a bump. Finally, two dents lie on edges with no common vertex, since dents occur on every second edge, and so they are disjoint by the second paragraph. The same holds for bumps. $\square$

Condition (G) holds well beyond $|h| < \delta/2$: at $p = 29/50$, an exact computation finds it up to $|h| \approx 0.685$.

Lemma 3.3.

- (1) For fixed p , condition (N) fails for only finitely many $|h| \in (0, \delta/2)$.
- (2) Let p and h be rational with $p(1-p) \neq h^2$. If

$$t = \frac{|h|}{p(1-p) - h^2} \notin \{0, 1, -1\},$$

then φ is not a rational multiple of π , and (N) holds.

Proof. (1) As $|h|$ increases, α and β increase, so φ is a continuous decreasing function of $|h|$. On $(0, \delta/2)$ it stays above $\varphi_0 = \varphi(\delta/2) > 0$. If (N) fails, then $\varphi = k\pi/(6j)$ with $1 \leq j \leq 2\pi/\varphi_0$ and $1 \leq k \leq 12$. These are finitely many values, and each is taken at most once.

(2) The addition formula gives $\tan(\alpha + \beta) = t$, a rational number. By Niven's theorem [Niv56, Chapter 3], the only rational values of the tangent at rational multiples of π are 0 and ± 1 . So $\alpha + \beta$, and hence $\varphi = \pi - (\alpha + \beta)$, is not a rational multiple of π . $\square$

3.3. The corners of X .

Lemma 3.4. Assume (G).

- (1) The angle of X at v_i is θ_i . In particular v_{10} is not a corner of X : the sides c_9v_{10} and $v_{10}c_{10}$ are collinear.
- (2) The angle of X at c_i is φ if Δ_i is a bump and $2\pi - \varphi$ if Δ_i is a dent.
- (3) Both sides of X at a blue vertex have length a , and both sides at a red vertex have length b .

So X is a polygon with 27 corners: the 13 vertices v_i with $i \neq 10$, seven apexes with angle φ and seven with angle $2\pi - \varphi$. Of its 27 sides, 26 have length a or b , and the side c_9c_{10} has length $2a$ and midpoint v_{10} .

Proof. (3) For even i , γ_i runs from the blue vertex v_i to c_i with length a , then to the red vertex v_{i+1} with length b . For odd i it runs from the red vertex v_i with length b , then to the blue vertex v_{i+1} with length a .

(1) Let ρ be the rotation about v_i that maps v_{i-1} to v_{i+1} , and let R be its linear part, which commutes with J . The point $E_{i-1}(u, w)$ lies at offset $(1-u)(v_{i-1} - v_i) - wJ(v_{i-1} - v_i)$ from v_i , so its image lies at offset $(1-u)(v_{i+1} - v_i) - wJ(v_{i+1} - v_i)$, and

$$\rho(E_{i-1}(u, w)) = E_i(1-u, -w).$$

Consecutive paths differ by the half turn $(u, w) \mapsto (1 - u, -w)$, so $\rho(\gamma_{i-1}) = \gamma_i$ and ρ maps the side $v_i c_{i-1}$ onto the side $v_i c_i$. At a corner of a simple anticlockwise polygon, the interior angle is the anticlockwise angle from the outgoing side to the reversed incoming side. For T at v_i these sides lie along the rays $v_i v_{i+1}$ and $v_i v_{i-1}$; for X , which is simple and anticlockwise by (G), along $v_i c_i$ and $v_i c_{i-1}$. The rotation ρ maps the second ray of each pair onto the first through the same angle. So the two interior angles are equal. At v_{10} the angle is π , so c_9, v_{10} and c_{10} are collinear.

(2) This is the apex angle of Δ_i , seen from inside a bump or from outside a dent.

The count of corners and sides follows from (1)–(3), since $|c_9 v_{10}| = |v_{10} c_{10}| = a$. $\square$

Lemma 3.4(1) says that X is an instance of the second construction of [SMKG24b, Lemma 2.1], in which the paths on consecutive edges are related by the rotation about their common vertex. That lemma assumes a smooth path; here the path is a triangle.

4. LOCAL STRUCTURE OF TILINGS BY X

Throughout this section, $X = X(p, h)$ satisfies (G) and (N), and $\mathcal{T}$ is a tiling by X . The *edge path* of a tile $S = gX$ through its apex gc_i is the image $g\gamma_i$: two sides, from one vertex gv_i to the next.

Lemma 4.1. *Let $x \in \mathbb{R}^2$.*

- (1) *If x is an apex of some tile, then exactly two tiles contain x . One has a bump apex at x , with angle φ , and the other a dent apex, with angle $2\pi - \varphi$. The two tiles have the same two boundary rays at x .*
- (2) *If x is an apex of no tile, then every tile containing x has angle $\pi/2, 2\pi/3, \pi, 4\pi/3$ or $3\pi/2$ at x . Moreover, no red corner of a tile lies in the interior of a side of another tile, and no blue corner of a tile coincides with a red corner of another.*

Proof. (1) Let m bump apexes and n dent apexes lie at x , and let the other angles at x sum to σ . By Lemma 3.4, σ is a multiple of $\pi/6$, and by Lemma 2.1

$$m\varphi + n(2\pi - \varphi) + \sigma = 2\pi.$$

Since $2\pi - \varphi > \pi$, $n \leq 1$. If $n = 0$, then $m \geq 1$ and $m\varphi = 2\pi - \sigma$ is a multiple of $\pi/6$ with $m \leq 2\pi/\varphi$, against (N). If $n = 1$, then $(m - 1)\varphi = -\sigma \leq 0$, so $m \leq 1$; the case $m = 0$ gives $\varphi = \sigma$, against (N). So $m = n = 1$ and $\sigma = 0$. The two sectors fill the full angle, so they share their boundary rays.

(2) The first claim follows from Lemma 3.4 and part (1). A corner of a tile whose angle is a multiple of $\pi/6$ never lies at an apex: by part (1), the only tiles containing an apex have angles φ and $2\pi - \varphi$ there, and by (N) neither is a multiple of $\pi/6$. So the remaining claims concern points that are apexes of no tile. An angle π plus a red angle leaves at most $\pi/3$, which is less than every available angle. A blue angle ($\pi/2$ or $3\pi/2$) plus a red angle ($2\pi/3$ or $4\pi/3$) leaves $5\pi/6, \pi/6$ or a negative amount, and neither $5\pi/6$ nor $\pi/6$ is a sum of available angles. $\square$

Lemma 4.2. *Let x be an apex of a tile A , and let B be the other tile containing x . Then A and B have the same hand, and their edge paths through x coincide, end points included.*

Proof. By Lemma 4.1(1), A and B share both boundary rays at x . Along each ray, each tile has a maximal straight piece of boundary starting at x , which we call its *run*. By Lemma 3.4, a run is one of:

- a side of length b , ending at a red corner (a *b-run*);
- a side of length a , ending at a blue corner; or the side $c_9 c_{10}$, of length $2a$, which passes v_{10} at distance a and ends at an apex (both *a-runs*).

Walk around each tile anticlockwise. On an unreflected tile, by Lemma 3.4(3), the walk passes γ_i as a -side then b -side when i is even, and as b -side then a -side when i is odd. Dents occur on the edges of one parity and bumps on the other, so dents and bumps are passed in opposite orders. A reflection reverses the order in which every edge path is passed and keeps dents as dents. So on a reflected tile, dents and bumps are also passed in opposite orders, each reversed.

At x each tile's sector lies to the left of its own walk. The two sectors are complementary, so the two walks pass the two rays in opposite orders. One of A and B has a dent at x and the other a bump.

Same hand. The edge paths are passed in opposite orders, and the walks run in opposite directions, so the two runs on each ray are of the same kind. The two b -runs end together at a red corner of each tile. The two a -runs both reach distance a at a blue corner or at v_{10} of each tile. So the two edge paths through x are the same set, and their end points are images of vertices of T in both tiles.

Opposite hands. Now the edge paths are passed in the same order, so each ray carries one a -run and one b -run. Fix a ray r , let N be the tile with the b -run on r and M the tile with the a -run, and let y be the red corner of N at distance b from x along r (Figure 3). We derive a contradiction in each case.

Case $a > b$. Then y lies in the interior of M 's run, a single side of M . This contradicts Lemma 4.1(2).

Case $a = b$. If M 's run is the side c_9c_{10} , then y lies in its interior. Otherwise M 's run ends at a blue corner of M at y . Both contradict Lemma 4.1(2).

Case $a < b$. Suppose first that M 's run is the side c_9c_{10} , ending at an apex z of M at distance $2a$ from x . If $2a < b$, then z lies in the interior of a side of N , against Lemma 4.1(1). If $2a > b$, then y lies in the interior of a side of M , against Lemma 4.1(2). If $2a = b$, then an apex and a red corner coincide, against Lemma 4.1(1).

Otherwise M 's run ends at a blue corner y' of M at distance $a < b$, in the interior of N 's side. The point y' is an apex of no tile, by Lemma 4.1(1). At y' , N has angle π . The angle of M at its blue corner y' is $\pi/2$ or $3\pi/2$, and at most π , so it is $\pi/2$. By Lemma 4.1(2), a single further tile C fills the remaining right angle, with a corner of angle $\pi/2$. That corner is a vertex of C with angle $\pi/2$, so it is blue. The sector of C at y' lies between the other side of M at y' and the continuation of r beyond y' . So C has a side along r , starting at y' and running away from x . By Lemma 3.4(3) it has length a , and since the angle of C at y' is not π , it ends at an apex z of C at distance $2a$ from x . Comparing $2a$ with b as in the previous paragraph, with C in place of M , gives a contradiction in each case. $\square$

5. PROOF OF THE MAIN THEOREM

Theorem 5.1. *Let $0 < p < 1$ and $h \neq 0$ satisfy (G) and (N). Then $X = X(p, h)$ is a strictly chiral aperiodic monotile:*

- (1) X tiles the plane;
- (2) every tiling by X is homochiral;
- (3) no tiling by X is periodic.

Moreover, X is a Spectre in the sense of [SMKG24b, Section 2].

Theorem 1.1 follows from Theorem 5.1, Proposition 3.2, Lemma 3.3(1) and Lemma 3.4.

Proof. Fix a tiling $\mathcal{T}$ by X .

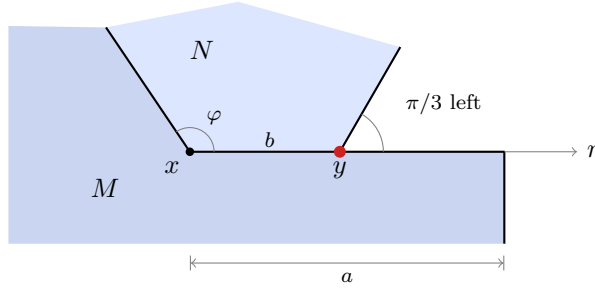


FIGURE 3. Two tiles of opposite hands at an apex x , in the case $a > b$. Here N has a bump apex at x and M a dent apex. Along the ray r , the b -run of N ends at a red corner y inside the a -run of M . The angle left at y is at most $\pi/3$, and no tile has so small an angle.

Step 1: edge paths match. Every edge path of every tile contains an apex. By Lemmas 4.1(1) and 4.2, each edge path of a tile S coincides with an edge path of exactly one other tile, which we call the *partner* of S along that path. The partner has the same hand as S . The triangle between the shared path and its chord is a bump of one of the two tiles and a dent of the other.

Step 2: one hand. Let $x \in \mathbb{R}^2$, and list the tiles containing x in cyclic order around x . Two consecutive tiles share a short boundary segment starting at x . It lies in an edge path of the first tile, whose partner lies across the segment and is therefore the second tile. So consecutive tiles have the same hand, and all tiles containing x have the same hand. By Lemma 2.1, a small disc about x meets only these tiles. So the points covered by unreflected tiles form an open set, and so do the points covered by reflected tiles. These sets are disjoint and cover the connected plane, so one of them is empty. This proves (2).

Step 3: straightening. For a tile $S \in \mathcal{T}$, let $\hat{S}$ be the polygon whose vertices, in boundary order, are the 13 corners of S whose angles are multiples of $\pi/6$, and the midpoint of the side of S whose two ends are apexes. By (N) the apex angles are not multiples of $\pi/6$, so $\hat{S}$ depends only on the set S , and by Lemma 3.4, $\hat{S} = gT$ whenever $S = gX$. The polygon $\hat{S}$ is S with each bump removed and each dent filled. By Step 1, each such triangle is a bump of one tile and a dent of its partner, so straightening moves it from the one to the other. Each triangle lies in the tile for which it is a bump, and the bumps of one tile are disjoint by (G), so the triangles have disjoint interiors. Hence $\hat{\mathcal{T}} = \{\hat{S} : S \in \mathcal{T}\}$ is a tiling by T , and it is homochiral by Step 2.

Step 4: no period. The map $S \mapsto \hat{S}$ commutes with isometries of the plane. If $\mathcal{T} + v = \mathcal{T}$, then $\hat{\mathcal{T}} + v = \hat{\mathcal{T}}$, and Theorem 2.2(3) gives $v = 0$. This proves (3).

Step 5: a tiling exists. Let $\{g_k T\}$ be a tiling by unreflected copies of T , which exists by Theorem 2.2(2). Two tiles $g_k T$ and $g_l T$ that share an edge run along it in opposite directions, and by Lemma 2.3 they agree on which end is blue. So the edge is $g_k e_i$ with i even in one tile, running from blue to red, and $g_l e_j$ with j odd in the other. The point $g_k E_i(u, w)$ equals $g_l E_j(1 - u, -w)$. Since γ^* is the half turn of γ , both tiles place the same path along the edge, with one triangle as a bump of one tile and a dent of the other. By (G), the dents of each $g_k X$ lie in $g_k T$ and are pairwise disjoint. So the sets $g_k X$ have disjoint interiors and cover the plane. This proves (1).

Spectre. The definition in [SMKG24b, Section 2] asks for four properties of X : every tiling by X is homochiral (Step 2); if $\{g_i X\}$ is a tiling then so is $\{g_i T\}$ (Step 3); if $\{g_i T\}$ is a homochiral tiling then so is $\{g_i X\}$ (Step 5, applied to either hand); and the two tilings have

the same points that lie in three or more tiles. For the last, straightening changes the tiles containing a point only inside the moved triangles. Apart from its base end points, each point of a triangle lies in at most two tiles, before and after. The base end points are vertices of both tiles sharing the edge, before and after. $\square$

6. EXAMPLES

Proof of Corollary 1.2. (1) Here $|h| = 0.26 < \delta/2 \approx 0.448$, so (G) holds by Proposition 3.2. Also $p(1-p) - h^2 = \frac{609-169}{2500} = \frac{440}{2500}$, so

$$t = \frac{13/50}{440/2500} = \frac{65}{44},$$

and (N) holds by Lemma 3.3(2). Numerically, $\varphi \approx 124.095^\circ$.

(2) Here $\alpha = \beta = \arctan(2|h|) < \pi/12$, and $|h| < \delta/2$, so (G) holds. Also $\varphi = \pi - 2\arctan(2|h|)$ lies strictly between $5\pi/6$ and π , so $2\pi/\varphi < 12/5$ and only $j = 1, 2$ need checking in (N). The angle φ is not a multiple of $\pi/6$. The angle 2φ lies strictly between $5\pi/3$ and 2π , so it is a multiple of $\pi/6$ only if $2\varphi = 11\pi/6$, that is, $|h| = \frac{1}{2} \tan \frac{\pi}{24}$. $\square$

The tile $X(29/50, 13/50)$ is the *Triangle* edge shape, in the *Single* arrangement, of the author's interactive tiling viewer at <https://drbuild.uk/tiles>, with peak position 0.58 and height 0.26. Figure 4 shows a patch of its tiling, built from a homochiral tiling by $\text{Tile}(1, 1)$ generated by the substitution rules of [SMKG24b, Figure 2.1 and Appendix A].

Computer check. The proofs above need no computer. As a check, the ancillary file `verify-polygonal-spectre.py` recomputes, in exact arithmetic over $\mathbb{Q}(\sqrt{3})$, the angles of $\text{Tile}(1, 1)$ and their colours, the value of δ , Lemma 3.4(1) and (3), the value of t , and condition (G) for any rational (p, h) . It uses no floating point in any test.

7. REMARKS AND OPEN QUESTIONS

Remark 7.1. Goucher's 28-edge polygon [Gou23] has unit sides meeting at turns of $\pm\pi/3$ at the apexes, so it is $X(1/2, \sqrt{3}/6)$ scaled by $\sqrt{3}$. It satisfies (G) but not (N): its apex angles are $2\pi/3$ and $4\pi/3$, the same as the red angles of $\text{Tile}(1, 1)$. The proof of Lemma 4.1(1) then fails, since an apex could meet a red corner. We do not know whether it is a Spectre.

Remark 7.2. Where (N) fails for a tile satisfying (G), the tile may still be a Spectre; the proof above does not decide it. In Corollary 1.2(2), the single excluded height $|h| = \frac{1}{2} \tan \frac{\pi}{24}$ is such a case.

Question 7.3. Does every non-straight piecewise-linear path, placed on the edges of $\text{Tile}(1, 1)$ by either construction of [SMKG24b, Lemma 2.1] so that the tile is a topological disc, give a Spectre? In particular, is the half-turn-symmetric zigzag $(0, 0), (q, k), (1 - q, -k), (1, 0)$ a Spectre for small $k \neq 0$?

Question 7.4. What is the least number of sides of a polygon that is a strictly chiral aperiodic monotile? The tiles here have 27 sides.

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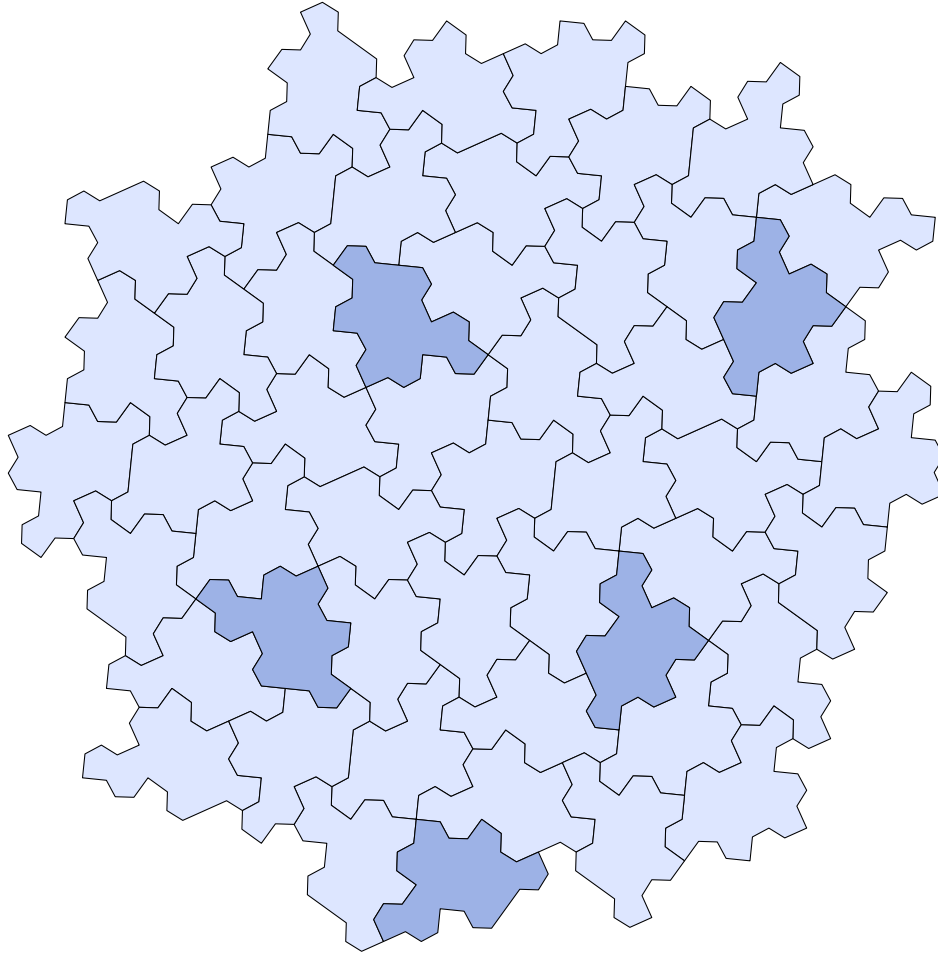


FIGURE 4. A patch of a tiling by $X(29/50, 13/50)$. Darker tiles are turned by an odd multiple of $\pi/6$ relative to the lighter ones.

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